Highlights

Rethinking Decoder Redundancy in Efficient 3D Medical Image Segmentation: A Characterization Study

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- Characterizes the spatial contribution of additional decoder computation in 3D segmentation.

- Establishes the distinction between decoder activity (where it acts) and net gain (whether it helps).

- Demonstrates net-neutral global voxel correctness alongside extreme boundary concentration ($K_{80} = 5\%$).

- Proves predictive uncertainty signals activity but fails to predict directional decoder utility.

Rethinking Decoder Redundancy in Efficient 3D Medical Image Segmentation: A Characterization Study

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Abstract

Modern 3D medical image segmentation architectures increasingly incorporate heavy decoders, assuming that dense multi-resolution decoding is universally necessary. Efficient alternatives simplify these stages based on global metrics, assuming static capacity is uniformly redundant. However, where and under what spatial conditions decoder computation actually contributes remains uncharacterized. In this work, we present a systematic characterization study of decoder computation in 3D medical image segmentation. By evaluating voxel-level prediction shifts between matched full and efficient decoders across representative architectures and datasets, we analyze the contribution of additional decoder computation across three dimensions: spatial heterogeneity, benefit concentration, and inference-time predictability. Our findings establish a core distinction: decoder activity measures where the decoder acts, whereas decoder gain measures whether those changes are beneficial. Globally, full decoders yield a net-neutral benefit as positive corrections are offset by negative degradations. Spatially, positive net gain is highly concentrated at organ boundaries, with 80% of positive benefit confined to 5% of the volume. Furthermore, while inference-time uncertainty reliably predicts where decoders act (Activity AUROC > 0.90), it fails to predict whether extra capacity will yield positive gain (Direction AUROC $\approx$ 0.55). This demonstrates that prediction difficulty and computational utility are related but fundamentally distinct quantities, providing essential empirical constraints for future adaptive decoder designs.

Keywords: 3D Medical Image Segmentation, Efficient Deep Learning, Uncertainty Quantification, Decoder Redundancy, Model Characterization

1. Introduction

Precise 3D medical image segmentation is essential for clinical tasks such as surgical planning, tumor monitoring, and radiation therapy. In deep learning architectures, encoder-decoder networks based on the U-Net paradigm [1, 2] remain the foundational standard. Recent research has primarily focused on developing highly expressive feature encoders, using vision transformers and large convolutional kernels [3, 4, 5, 6]. In parallel, modern 3D architectures retain heavy, multi-resolution decoders equipped with complex skip-fusion and refinement modules. This architectural trajectory assumes that dense decoder capacity is universally necessary across the image volume.

However, standard evaluation paradigms assess decoder designs almost exclusively through aggregate metrics such as the global Dice similarity coefficient or Hausdorff distance. While these macroscopic metrics summarize overall performance, they obscure the spatial distribution of prediction changes. Consequently, a basic scientific question remains open:

How does additional decoder computation actually contribute to 3D medical image segmentation?

To address this question, we present a characterization study of decoder computation in 3D segmentation models. Rather than introducing a new network architecture, we advocate computation characterization as a research approach complementary to architecture design. We structure our evaluation across two distinct experimental levels:

1. **End-to-End Model Comparison**: We evaluate matched full and efficient decoder pairs across standard end-to-end training settings to test whether spatial characterization patterns generalize across representative 3D backbones and clinical datasets.
2. **Decoder-Isolated Attribution Analysis**: Building on frozen shared-encoder factorials (derived from EffiDec3D [7]), we freeze feature extractor weights to strictly isolate decoder capacity interventions (channel reduction and stage removal), ensuring controlled causal attribution of prediction shifts (Figure 1; see Supplementary Figures S1 and S2 for corresponding SwinUNETR and MedNeXt-M-K3 qualitative visualizations).

We focus specifically on the decoder pathway as a well-defined subsystem responsible for spatial resolution reconstruction and multi-scale feature fusion. Skip-feature fusion is treated as part of the decoder pathway, whereas encoder representation dynamics present separate research questions requiring distinct experimental controls. Unlike efficient architecture papers that optimize for speed, our primary objective is to understand computation itself.

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We characterize the contribution of additional decoder computation across three complementary dimensions: (i) spatial heterogeneity, (ii) benefit concentration, and (iii) inference-time predictability.

The remainder of this paper is organized as follows: Section 2 reviews related literature and highlights key gaps. Section 3 describes our evaluation framework and metrics. Section 4 details our experimental results. Section 5 discusses underlying mechanisms and practical implications, and Section 6 concludes the paper.

2. Related Work

2.1. Decoder Design in 3D Medical Image Segmentation

The encoder-decoder architecture introduced by U-Net [1, 2] and V-Net [8] remains the central design pattern for 3D medical image segmentation. Frameworks such as nnU-Net [9] demonstrated the effectiveness of standardized training protocols combined with deep decoding pathways. Subsequent developments incorporated Transformer backbones and large-kernel convolutions to capture long-range contextual dependencies (e.g., TransUNet [10], UNETR [3], Swin UNETR [4], 3D UX-Net [5], and MedNeXt [6]). These modern architectures emphasize expressive feature extraction while maintaining dense multi-scale decoders. However, these designs are evaluated primarily through final segmentation accuracy and overall parameter count; how decoder capacity operates spatially across volume regions remains unexamined.

2.2. Efficient Segmentation Architecture Design

To mitigate the high computational cost of 3D volumetric convolutions, recent works have explored lightweight decoder designs. EffiDec3D [7] introduced systematic decoder pruning by reducing channel dimensions and eliminating high-resolution decoding stages, establishing trade-offs between parameter scale and Dice performance. Similarly, EfficientMedNeXt [11] simplified decoding stages before introducing residual blocks. Other lightweight models achieve efficiency through hybrid transformer-convolution encoders, attention pruning, or custom lightweight modules (e.g., SegFormer [12], UNETR++ [13], Slim UNETR [14]). Nevertheless, efficient architecture studies infer redundancy indirectly from global performance curves, leaving unexamined whether the eliminated computation was spatially uniform or concentrated in specific anatomical sub-regions.

2.3. Model Characterization and Uncertainty Quantification

Model characterization studies in medical imaging have focused on uncertainty estimation, calibration, failure detection, and quality assessment using techniques such as Monte Carlo Dropout [15] and Deep Ensembles [16] (e.g., Jungo & Reyes [17], recent CMIG studies [18, 19], and comprehensive reviews [20]). These works analyze how uncertainty maps correlate with segmentation errors or clinician confidence. However, existing characterization frameworks assess general prediction uncertainty rather than the marginal spatial utility of decoder computation. The relationship between spatial uncertainty and decoder capacity remains unexplored.

3. Methods: A Characterization Framework

To understand the spatial and functional contribution of decoder computation, we establish a controlled observation framework. We evaluate matched pairs of full and efficient decoders as experimental probes under two complementary regimes: (i) *end-to-end model comparison* across independently trained networks to assess real-world generalization, and (ii) *decoder-isolated attribution analysis* using frozen shared encoders to isolate decoder capacity interventions.

3.1. Controlled Decoder Comparison

Let v denote a voxel within a 3D medical image volume V. We compare two decoder configurations:

- **Full Decoder** (E_0): The standard dense, multi-stage high-resolution decoder of the respective backbone architecture.
- **Efficient Decoder** (E_1): A parameter-matched efficient decoder following the EffiDec3D specification [7], utilizing channel reduction and stage removal with concatenation skip aggregation.

Let $y(v)$ be the ground truth label, $\hat{y}_f(v)$ be the prediction from the full decoder, and $\hat{y}_e(v)$ be the prediction from the efficient decoder. Interventions occur strictly within the decoder pathway (upsampling, skip fusion, and refinement).

3.2. Decoder-Induced Corrections and Degradations

Comparing predictions between the efficient and full decoders at each voxel yields three possible outcomes: the full decoder corrects an error, introduces an error, or leaves prediction correctness unchanged:

- **Positive Correction** ($P(v)$): $P(v) = 1[\hat{y}_e(v) \neq y(v) \wedge \hat{y}_f(v) = y(v)]$.
- **Negative Degradation** ($N(v)$): $N(v) = 1[\hat{y}_e(v) = y(v) \wedge \hat{y}_f(v) \neq y(v)]$.

We define two fundamental analytical quantities:

- **Decoder Activity** ($A(v) = P(v) + N(v)$): Indicates where additional decoder capacity alters prediction correctness, regardless of direction.
- **Decoder Net Gain** ($G(v) = P(v) - N(v)$): Distinguishes beneficial corrections from harmful degradations.

Decoder activity measures where the decoder acts, whereas decoder gain measures whether those changes are beneficial or harmful.

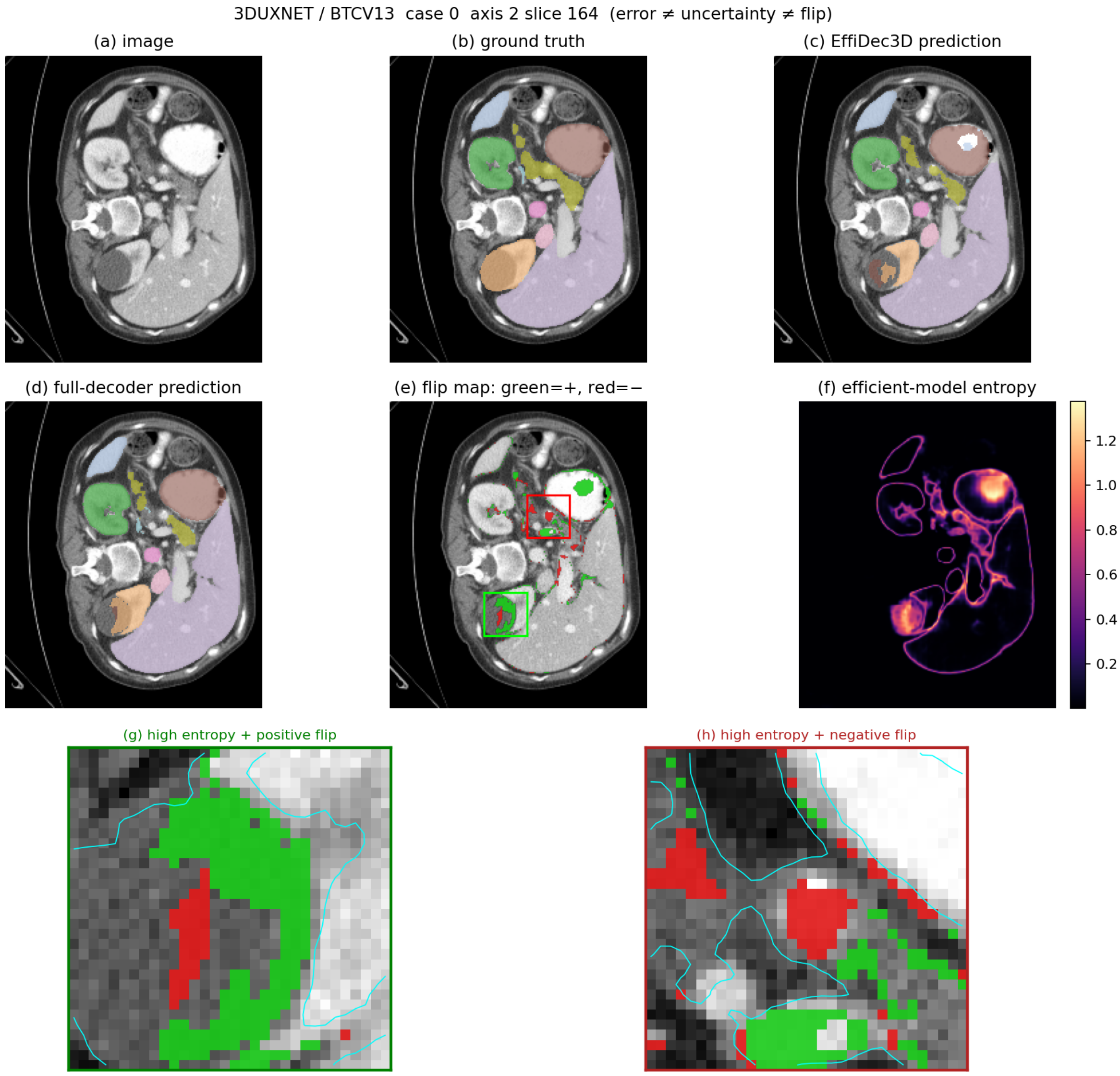


Figure 1: Motivation: Decoder computation produces spatially localized prediction flips. While the full decoder corrects certain errors of the efficient baseline (positive flips), it also alters previously correct predictions (negative flips) near high-entropy organ boundaries, demonstrating non-uniform spatial contribution (see Supplementary Figures S1 and S2 for SwinUNETR and MedNeXt-M-K3).

Table 1: Overview of characterization objectives and evaluation measures.

Analysis Objective	**Evaluation Measure**	**Question Answered**
Spatial effect	Activity $A(v)$	Where does additional decoder computation change predictions?
Directional effect	Net Gain $G(v)$	Where does the decoder improve correctness overall?
Concentration	Oracle K_{80}	How spatially concentrated is positive net gain?
Activity predictability	Location AUROC	Can cheap signals detect where the decoder acts?
Gain predictability	Direction AUROC	Can cheap signals distinguish beneficial from harmful changes?
Practical recovery	Routing Gain	Does selective execution improve segmentation metrics?

3.3. Spatial Characterization of Decoder Effects

We analyze Activity $A(v)$ and Net Gain $G(v)$ across two spatial dimensions:

1. **Boundary-Distance Stratification**: Distance to the nearest organ boundary using a multi-class union foreground mask $\bigcup_c (y(v) = c \cup \hat{y}_e(v) = c \cup \hat{y}_f(v) = c)$ [21].
2. **Anatomical Structure Breakdown**: Aggregated $A(v)$ and $G(v)$ across individual organ structures.

3.4. Spatial Concentration and Oracle Opportunity

We partition the 3D volume into spatial sub-blocks b_r and compute block net gain:

$$G(r) = \sum_{v \in b_r} G(v) \tag{1}$$

Sorting blocks by $G(r)$, the fraction of positive gain recovered by the top $k\%$ of blocks is defined by the coverage curve:

$$Coverage(k) = \frac{\sum_{r=1}^{k\% \cdot |B|} \max(G(r), 0)}{\sum_{r=1}^{|B|} \max(G(r), 0)} \tag{2}$$

We record K_{80}, defined as the smallest fraction of spatial blocks required by a perfect selector to recover 80% of the total positive decoder gain. A selective hybrid prediction $\hat{y}_k(v)$ executes the full decoder on selected blocks S_k and the efficient baseline elsewhere:

$$\hat{y}_k(v) = \begin{cases} \hat{y}_f(v), & \text{if } v \in S_k \\ \hat{y}_e(v), & \text{otherwise} \end{cases} \tag{3}$$

3.5. Predictability and Practical Routing Gain

We compute predictive Shannon entropy $H(v)$ from the efficient baseline outputs [24, 15]:

$$H(v) = -\sum_{c=1}^{C} p_c(v) \log p_c(v) \tag{4}$$

where $p_c(v)$ represents the softmax probability for class c at voxel v. Using $H(v)$, we evaluate three predictability measures:

1. **Location AUROC**: Discrimination of active blocks ($A(r) > 0$) vs. inactive blocks ($A(r) = 0$) [25].
2. **Direction AUROC**: Discrimination of net positive blocks ($G(r) > 0$) vs. net negative blocks ($G(r) < 0$) [19].
3. **Practical Routing Gain**: Segmentation metric recovery (Dice/HD95) achieved by entropy-guided selection over the efficient baseline E_1, compared against a random selection baseline under identical spatial budgets.

4. Experiments and Results

4.1. Computational and Accuracy Trade-offs

We evaluate matched decoder pairs across three 3D medical image segmentation backbones: a large-kernel CNN (3D UX-Net [5]), a vision transformer (SwinUNETR [4]), and a ConvNeXt-style network (MedNeXt-M-K3 [6]). Benchmarks are conducted on the BTCV multi-organ CT dataset (13 organs) [27] and the FeTA 2021 fetal brain MRI dataset [28].

Baseline Comparison: Table 2 summarizes the aggregate computational complexity and segmentation performance under standard end-to-end model training. For each backbone, we evaluate the standard Full decoder (E_0) against an Efficient baseline (E_1) implementing the parameter reduction strategy of EffiDec3D [7]. Across all three backbones, E_1 achieves substantial FLOP reductions (up to 91.4% in 3D UX-Net) while retaining competitive Mean Dice scores (e.g., MedNeXt-M-K3 E_0 achieves 83.74% Mean Dice vs. 81.47% for E_1, while E_1 reduces inference latency from 40.9 ms to 18.7 ms).

Factorial Factor Analysis: In our decoder-isolated attribution analysis on 3D UX-Net with frozen shared encoders (Table 3), we separate channel reduction (C_{dec}) from stage removal (RF). Reducing channels from 384 to 48 (C_{48}, RF_1) compresses parameters from 46.72M to 2.24M while maintaining a 79.80% Mean Dice. Omitting the high-resolution stage (C_{384}, RF_2) reduces latency to 16.1 ms. Combining both (C_{48}, RF_2) yields the E_1 configuration (1.84M params, 49.49 GMac).

Aggregate metrics confirm the overall efficiency-accuracy trade-off, but they do not reveal where additional decoder computation acts or helps across volumetric space.

4.2. Spatial Heterogeneity of Decoder Contribution

4.2.1. Global Correction-Degradation Balance

Evaluating voxel-level prediction shifts between matched E_0 and E_1 decoders reveals that positive corrections $P(v)$ and negative degradations $N(v)$ are comparable in magnitude across backbones. The subject-mean net gain R_{net} remains small ($< 0.1\%$) across all models: +0.046% (95% CI: $[-0.020\%, +0.110\%]$) for 3D UX-Net, -0.0008% (95% CI: $[-0.044\%, +0.039\%]$) for SwinUNETR, and +0.085% (95% CI: $[+0.024\%, +0.152\%]$) for MedNeXt-M-K3 (Figure 2, left column). While 3D UX-Net and SwinUNETR exhibit global confidence intervals crossing zero, MedNeXt-M-K3 shows a small, statistically significant positive shift. Overall, positive voxel corrections are largely counterbalanced by negative degradations across global volumes.

4.2.2. Boundary Localization of Decoder Activity

Spatial stratification reveals that decoder activity $A(v) = P(v) + N(v)$ is concentrated near anatomical boundaries. Error rates in boundary regions are 3.93× higher than in organ interiors for 3D UX-Net and 4.67× higher for MedNeXt-M-K3. Figure 2 (right column) shows that both positive corrections and negative degradations peak sharply within a 0–1 voxel boundary band, demonstrating that decoder capacity acts primarily at spatial transitions.

4.2.3. Factorial Attribution of Channel vs. Stage Interventions

To isolate the specific architectural mechanisms driving boundary localization, we examine the pairwise spatial flip distributions from our frozen shared-encoder factorial ablation on 3D UX-Net:

- **High-Resolution Stage Removal ($RF_1 \rightarrow RF_2$)**: Removing the high-resolution stage accounts for the majority of compute reduction (657.8 GMac → 319.1 GMac). Crucially, the net benefit of retaining this high-resolution stage is sharply boundary-concentrated, yielding a net positive correction rate of +0.033% in the 0–1 voxel boundary band and +0.031% in the 1–2 voxel band (both 95% CIs excluding zero), decaying to negative net values (-0.011%) in organ interiors (> 4 voxels). This establishes that high-resolution decoding stages specifically drive boundary refinement.

Table 2: Quantitative comparison of Full (E_0) vs. Efficient (E_1) decoders on the BTCV 13-organ dataset across three backbones.

Architecture	Variant	Params (M) ↓	FLOPs (GMac) ↓	Latency (ms) ↓	Mean Dice (%) ↑	Mean HD95 (mm) ↓
3D UX-Net	Full (E_0)	53.01	578.74	40.6	79.18	9.04
	Effi (E_1)	3.16	49.49	8.0	77.73	11.82
SwinUNETR	Full (E_0)	62.19	308.24	37.0	80.48	8.88
	Effi (E_1)	11.21	55.84	16.7	79.49	8.90
MedNeXt-M-K3	Full (E_0)	39.32	230.87	40.9	83.74	5.77
	Effi (E_1)	12.95	106.91	18.7	81.47	6.60

Table 3: 2×2 Factorial analysis of decoder channel reduction (C_{dec}) and resolution scaling (RF) on 3D UX-Net (BTCV dataset).

Configuration	C_{dec}	RF	Params (M)	FLOPs (GMac)	Latency (ms)	Mean Dice (%)
Full (C_{384}, RF_1)	384	1	46.72	657.76	43.3	80.44
Chan-Only (C_{48}, RF_1)	48	1	2.24	388.15	35.2	79.80
Res-Only (C_{384}, RF_2)	384	2	46.32	319.10	16.1	77.97
Effi (C_{48}, RF_2)	48	2	1.84	49.49	8.0	78.35

- **Channel Width Reduction** ($C_{384} \rightarrow C_{48}$): Reducing decoder channels compresses parameter count by 95% (46.7M $\rightarrow$ 2.2M) while exerting a spatially diffuse and net-neutral effect ($R_{net} = +0.027\%$, 95% CI: $[-0.027\%, +0.083\%]$). At the 0–1 voxel boundary, channel reduction produces a slight negative net rate (−0.020%), confirming that channel width provides diffuse representation capacity without driving localized boundary resolution.

4.2.4. Anatomical Volatility

Anatomical structure analysis (Figure 3) demonstrates that small, morphologically complex organs exhibit high flip volatility. The left and right adrenal glands show positive correction rates of 16.2%/16.9% in 3D UX-Net, 14.4%/14.8% in SwinUNETR, and 18.3%/10.4% in MedNeXt-M-K3. Conversely, large structures like the liver remain highly stable (< 3% Activity).

Extra decoder computation is largely inactive in smooth organ interiors, but frequently alters predictions near boundaries and difficult structures.

4.3. Spatial Concentration of Decoder Benefit

Figure 4 presents oracle coverage curves evaluating the spatial concentration of positive net gain $G(r)$. Across all three architectures, an oracle selecting the top 5% of blocks recovers 80% of the full decoder's total positive net gain ($K_{80} = 5\%$). At a 10% volume budget, oracle coverage reaches 99.98%, reaching 100% at 20%.

The top 5% of spatial blocks account for 80% of the observed positive net gain ($K_{80} = 5\%$), indicating strong spatial concentration of beneficial decoder effects. This does not imply that the remaining 95% of blocks are entirely inactive or dispensable, but rather that high-capacity decoder computation yields diminishing net returns outside this core subset.

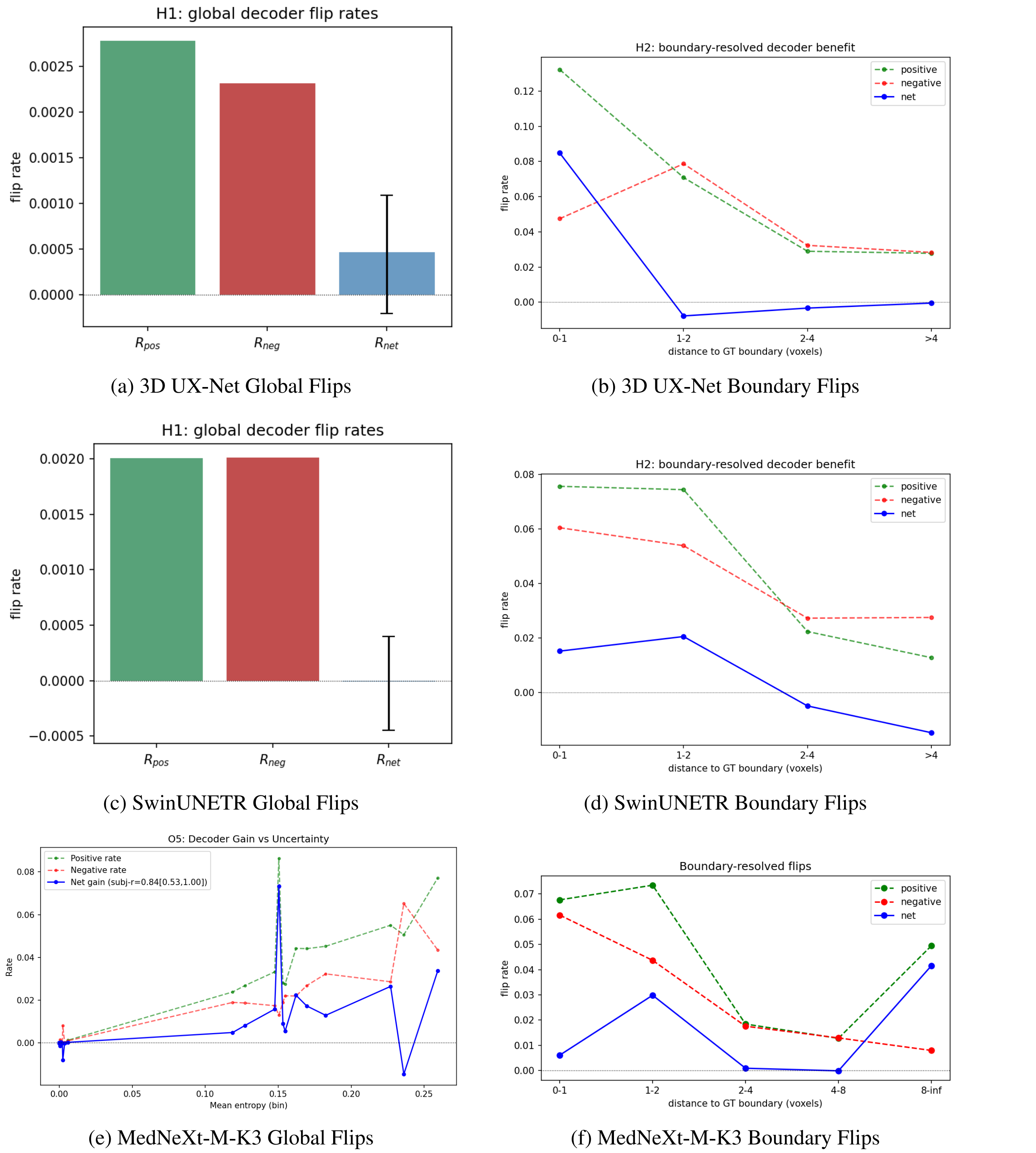
4.4. Predictability of Decoder Activity and Gain

We test whether cheap inference-time signals (predictive entropy) can locate these beneficial spatial regions prior to full decoder execution. Figure 5 shows a sharp divergence between predicting Activity and predicting Net Gain:

- **Activity Prediction (Flip Location AUROC)**: Predictive entropy reliably identifies active blocks ($A(r) > 0$), achieving high location AUROCs of 0.903 for 3D UX-Net, 0.912 for SwinUNETR, and 0.931 for MedNeXt-M-K3.

- **Directional Utility Prediction (Direction AUROC)**: Predictive entropy fails to distinguish positive corrections from negative degradations ($G(r) > 0$ vs. $G(r) < 0$), yielding direction AUROCs near random chance (0.592 for 3D UX-Net, 0.561 for SwinUNETR, 0.545 for MedNeXt-M-K3).

Entropy-guided routing provides architecture-dependent and incomplete recovery of the oracle opportunity: while entropy selection yields a small, statistically significant improvement over random selection at a 20% budget in 3D UX-Net, it fails to achieve significant gains in SwinUNETR and MedNeXt-M-K3. Overall, simple scalar entropy gating fails to reliably recover the theoretical oracle opportunity across backbones.



(a) 3D UX-Net Global Flips

(b) 3D UX-Net Boundary Flips

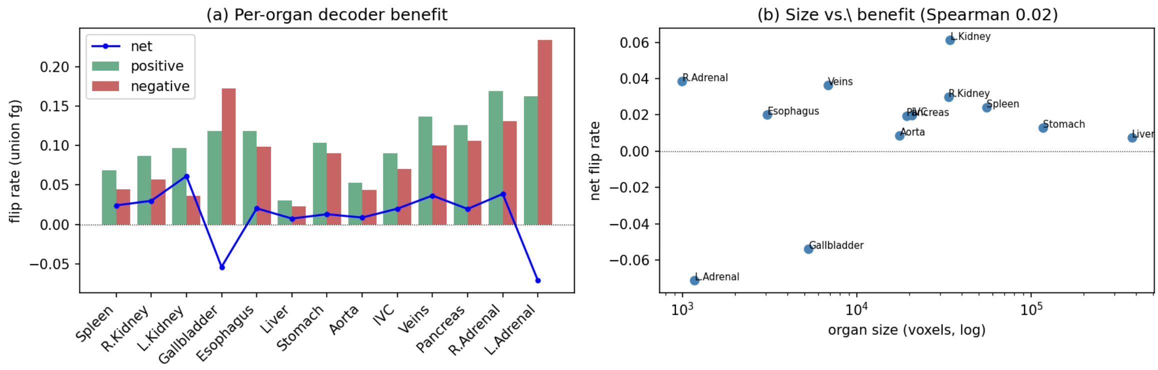
(c) SwinUNETR Global Flips

(d) SwinUNETR Boundary Flips

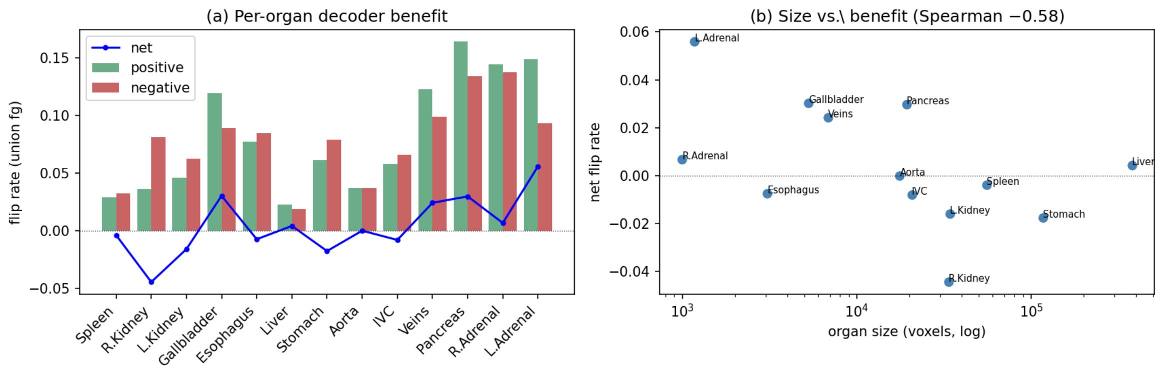
(e) MedNeXt-M-K3 Global Flips

(f) MedNeXt-M-K3 Boundary Flips

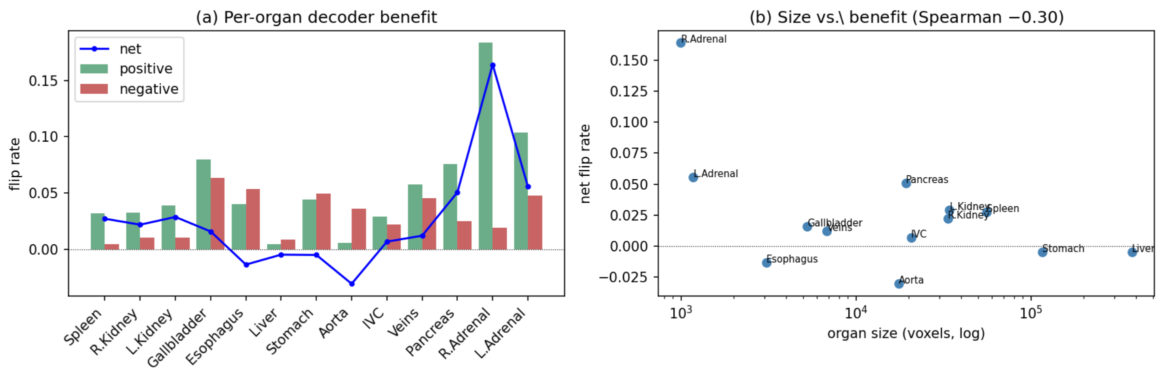
Figure 2: Voxel-level net gain and spatial boundary concentration across architectures. Left Column: Global positive corrections, negative degradations, and net gain (R_{net} < 0.1%). Right Column: Activity and net gain as a function of distance to organ boundaries.



(a) 3D UX-Net Organ-Level Flips

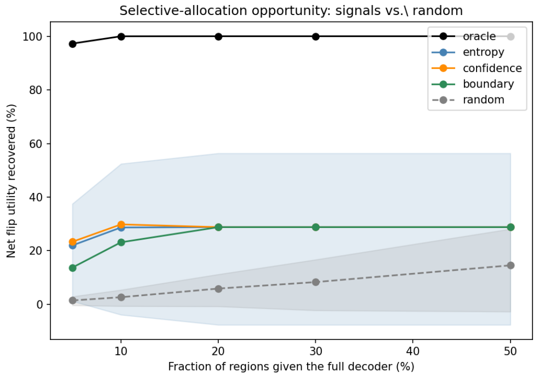


(b) SwinUNETR Organ-Level Flips

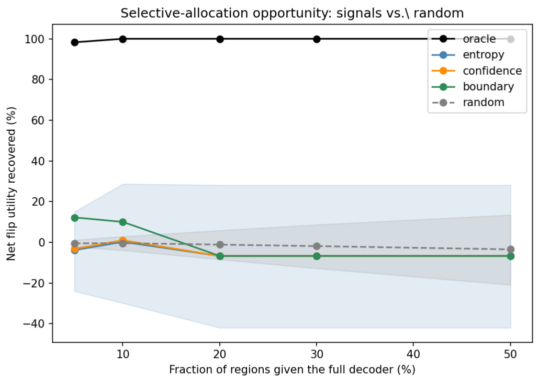


(c) MedNeXt-M-K3 Organ-Level Flips

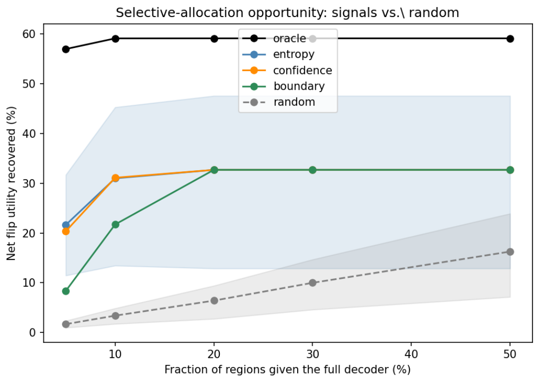
Figure 3: Organ-level flip breakdown across backbones. Complex and small anatomical structures show high prediction volatility, whereas large organs remain stable.



(a) 3D UX-Net Concentration Curve

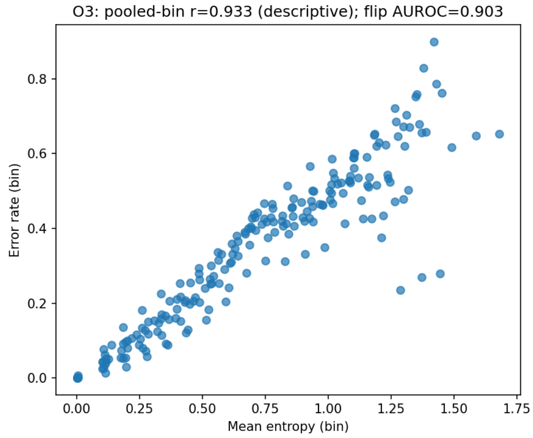


(b) SwinUNETR Concentration Curve

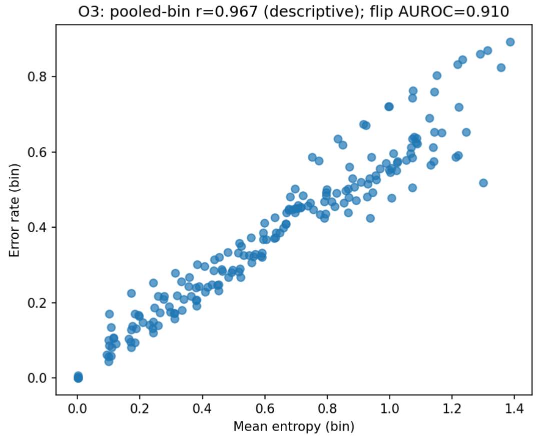


(c) MedNeXt-M-K3 Concentration Curve

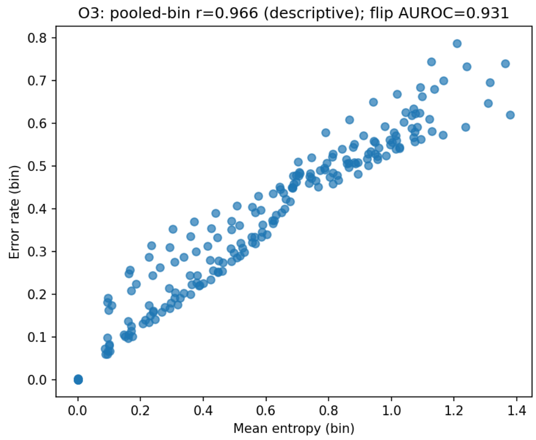
Figure 4: Oracle concentration curves showing that 80% of positive net gain (K_{80}) is concentrated in 5% of spatial blocks across all three backbones.



(a) 3D UX-Net Predictability



(b) SwinUNETR Predictability



(c) MedNeXt-M-K3 Predictability

Figure 5: Inference-time predictability: Predictive entropy reliably identifies active blocks (Location AUROC > 0.90) but cannot distinguish positive corrections from negative degradations (Direction AUROC ≈ 0.55).

5. Discussion

This characterization study shifts the evaluation of decoder efficiency from empirical FLOPs reduction toward understanding spatial computation dynamics. This section synthesizes the underlying mechanisms and distills design directions for future adaptive 3D segmentation decoders.

5.1. Why Aggregate Metrics Hide Decoder Behavior

Standard evaluation paradigms assess segmentation models using global metrics like Dice or Hausdorff distance. These macroscopic metrics average positive corrections and negative degradations across the entire volume, concealing localized micro-dynamics. Furthermore, because homogeneous organ interiors dominate the voxel count, net boundary improvements are diluted by large inactive regions. This explains why conventional evaluation frameworks have historically encouraged uniform decoder expansion despite underlying spatial redundancy.

5.2. Spatial Concentration Creates an Opportunity, Not a Solution

Our findings demonstrate that positive decoder net gain is strongly concentrated ($K_{80} = 5\%$). However, it is essential to distinguish *intrinsic spatial concentration* from *practical deployable recoverability*. Oracle concentration proves that selective decoding is theoretically worthwhile under an ideal selector. But realizing these computational savings requires identifying beneficial regions prior to executing the full decoder. Concentration illustrates the theoretical opportunity; it does not solve the routing problem.

5.3. Uncertainty Is an Activity Signal, Not a Utility Signal

Why does predictive entropy predict flip locations ($>$ 0.90 AUROC) but fail to predict directional improvement ($\approx$ 0.55 AUROC)? Entropy measures epistemic task difficulty and spatial ambiguity, effectively locating boundary regions where predictions are unstable. However, entropy indicates region difficulty rather than whether a static full decoder possesses the correct inductive bias to fix the error. Because efficient baseline decoders are already competitive, executing additional decoder capacity on uncertain voxels is equally likely to degrade a correct prediction as it is to fix an error. *Prediction difficulty and computational utility are related but fundamentally distinct quantities.*

5.4. Design Directions for Adaptive Decoders

Based on these empirical findings, we distill three evidence-backed directions for designing future adaptive decoders:

1. **Explore Spatially Selective Refinement**: The concentration results motivate sparse or region-specific high-resolution decoding rather than uniformly allocating maximum decoder capacity across the entire volumetric grid.
2. **Predict Expected Gain Rather Than Uncertainty Alone**: Difficulty signals identify where predictions are unstable, but predicting directional improvement requires routing modules designed to estimate expected net gain from extra computation.
3. **Couple Routing More Closely with Refinement**: The limited performance of scalar post-hoc signals motivates exploring learned utility predictors or jointly optimized routing and refinement modules trained end-to-end with directional supervision.

6. Conclusion

This paper presented a characterization study of decoder computation in 3D medical image segmentation architectures. By evaluating voxel-level prediction shifts between matched full and efficient decoders, we established a fundamental distinction between decoder activity (where computation alters predictions) and decoder net gain (where computation improves overall correctness). Our results show that while useful decoder benefit is highly concentrated at organ boundaries ($K_{80} = 5\%$), global net gain remains net-neutral as positive corrections are offset by negative degradations. Furthermore, we showed that inference-time uncertainty signals reliably locate decoder activity but cannot predict directional decoder utility.

These findings demonstrate that prediction difficulty and computational utility are fundamentally distinct quantities, revealing the inherent limits of static decoding and naive uncertainty gating. Future work should focus on utility-aware dynamic decoders that jointly optimize spatial routing and feature refinement.

CRediT authorship contribution statement

Author One: Conceptualization, Methodology, Writing – original draft. **Author Two**: Data curation, Software. **Author Three**: Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that me could have appeared to influence the work reported in this paper.

Data availability

The data used in this study (BTCV/Synapse, FeTA 2021) are publicly available.

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Supplementary Material: Qualitative Visualizations for SwinUNETR and MedNeXt-M-K3

This section provides supplementary qualitative visualizations for SwinUNETR and MedNeXt-M-K3 backbones, complementing the 3D UX-Net qualitative motivation example in Figure 1 of the main manuscript.

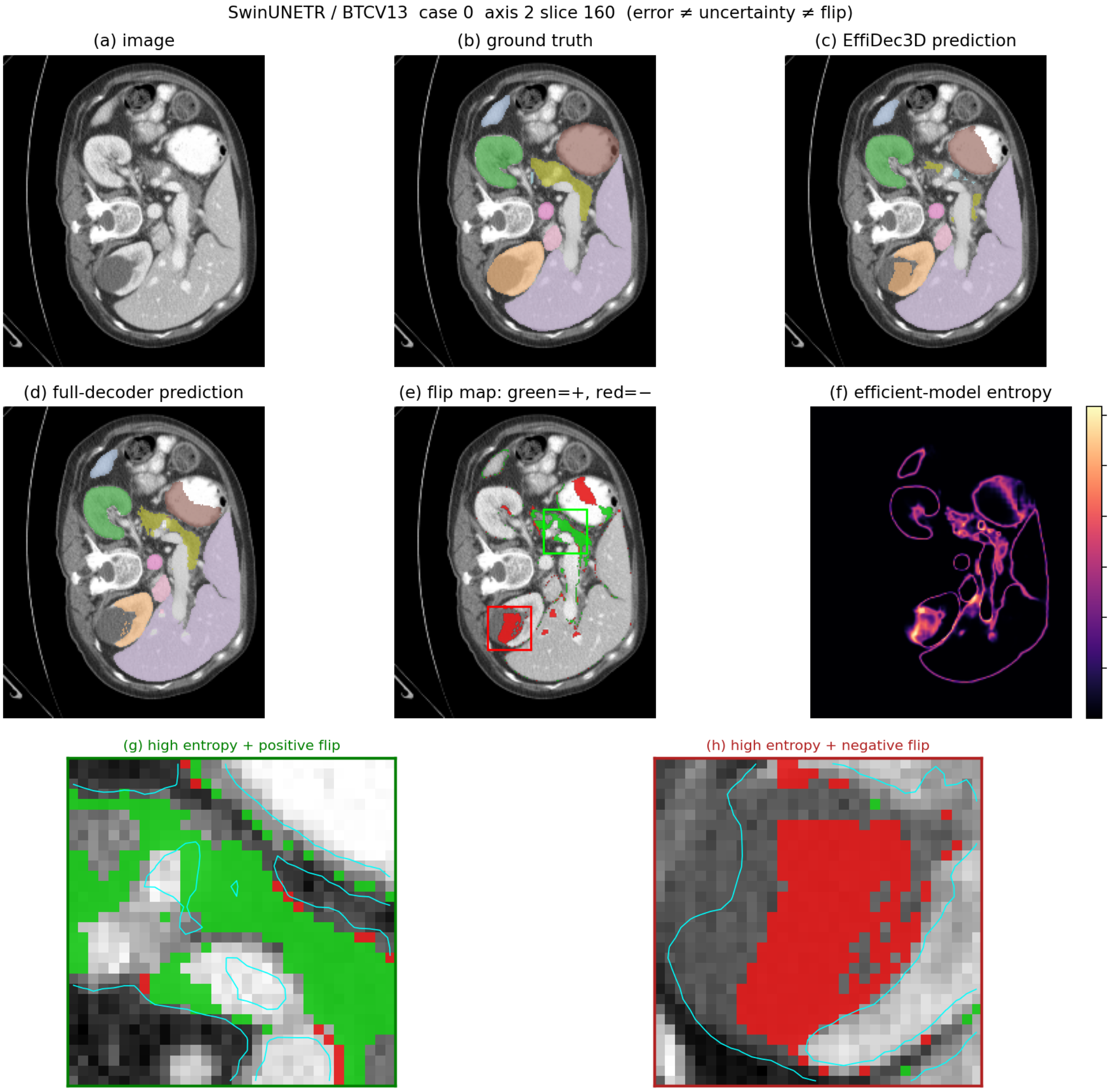

Figure S1: Supplementary Figure S1: Qualitative motivation visualization for SwinUNETR (E_0 vs. E_1) on BTCV CT multi-organ segmentation. Voxel-level prediction flips (positive corrections and negative degradations) are tightly localized to high-entropy organ boundaries.

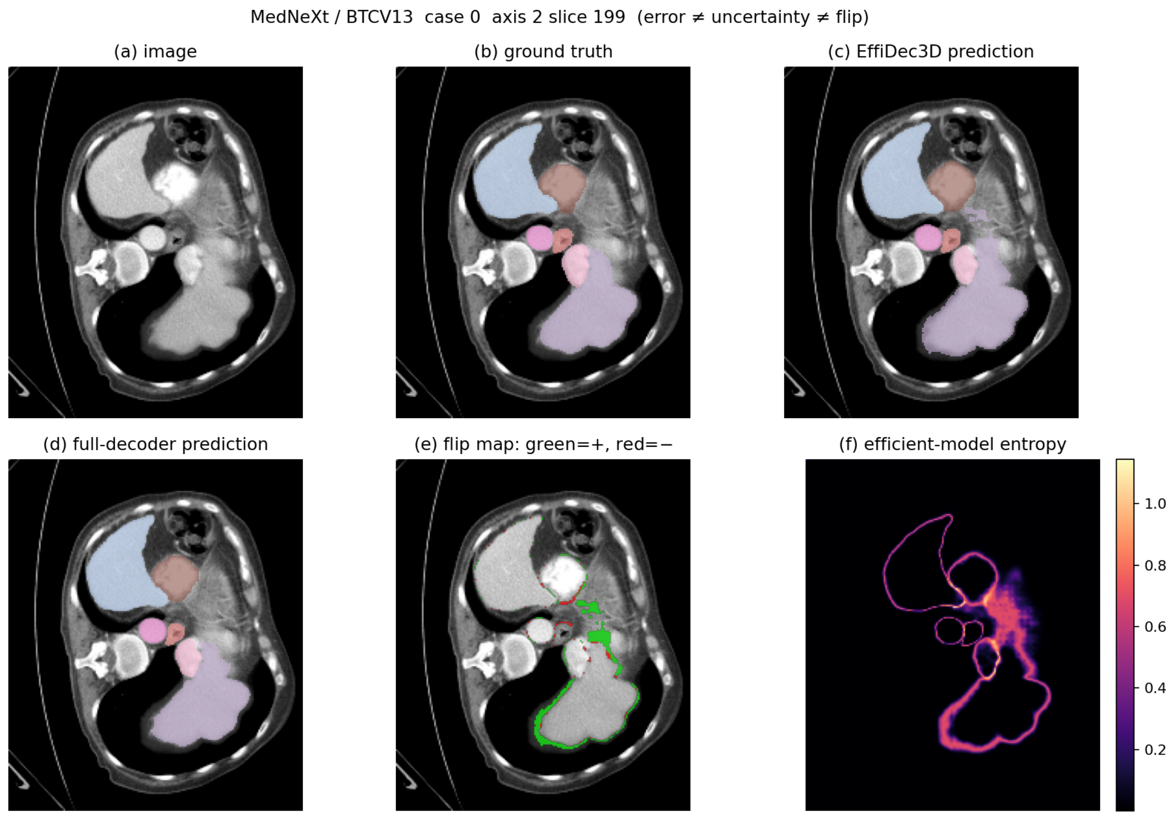

Figure S2: Supplementary Figure S2: Qualitative motivation visualization for MedNeXt-M-K3 (E_0 vs. E_1) on BTCV CT multi-organ segmentation. Consistent with 3D UX-Net and SwinUNETR, decoder-induced flips are concentrated within narrow boundary bands.